

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

December 2012

EE-414 Power System Operations

Date: 08.12.2012, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Answers the following questions.

[07]

- 1 Which are the different approaches used for load forecasting?
- 2 A Markov process with discrete index is called _____.
- 3 Define the term Unbundling.
- 4 What are the characteristics of a vertically integrated electric industry?
- 5 Write down the equality and inequality constraints of a economic dispatch problem.
- 6 $1 \text{ MBtu} = \text{_____ kwh.}$
- 7 Why reactive forecast is difficult than active load forecast?

Q - 2 (a) What is two state fluctuating environment? Explain Markov application to single repairable component operating in two state fluctuating environment. [07]

(b) A circuit is formed by 4 components A, 2 component B and 3 components C all connected in series. The failure rates of component A, B, C are 3×10^{-3} , 5×10^{-3} and 6×10^{-3} failures/hour. Find (a) MTTF (b) System reliability for an operating time of 12 hours. Assume exponential distribution. [03]

(c) Explain load forecasting methodology and estimation of average and trend terms. [04]

OR

Q - 2 (a) A radial system consist of a transformer and a feeder. The feeder is 5 km long. The failure rates for transformer and feeder are 1.5 failure/year and 0.8 failures/km/year respectively. The repair times for transformer and feeder are 8 hours and 6 hours per outage. Find (a) Failure rate of system (b) total outage time per year (c) down time per outage. [03]

(b) Draw typical bath tub curve. Explain the three regions of the curve. For which region are reliability calculations valid? Why? [07]

(c) What do you mean by load forecasting? Explain nature of forecast, lead times and [04]

applications.

- Q - 3 (a) Explain in detail the term 'Qualitative screening Analysis', 'Maximum technical potential Analysis'. [07]
- (b) What is the use of performance index? Explain IP1Q method in detail with flow chart. [07]

OR

- Q - 3 (a) What is the role of load management in DSM? How are load control techniques classified? Discuss each of the techniques. [07]
- (b) What is the purpose of contingency analysis? Explain Contingency analysis in detail. [07]

SECTION - II

- Q - 4 (a) What are the different entities in deregulated environment? [05]
- (b) Explain the function of Sensor, Transmitter, Transducer and Receiver. [02]
- Q - 5 (a) Give definition of the following terms:-(1)Man Machine Interface, (2)Modem, (3)LAN, (4)WAN, (5) Protocol (6)Remote terminal units, (7)Router. [07]
- (b) What do you mean by deregulation? Explain structure of deregulated industry. [05]
- (c) What are the conditions to network become observable? [02]

OR

- Q - 5 (a) Define the term Distribution Automation. Explain the primary functions of DA. [07]
- (b) How can we identify bad data from measurement matrix? [02]
- (c) Write short notes on Ancillary services. [05]
- Q - 6 (a) Explain Lambda-Iteration method of Economic dispatch problem in detail. [07]
- (b) Estimate two random variables and estimation error by weighted least square estimation method for a given measurement vector Z, H and Weight(W) as follows:-

$$Z = \begin{bmatrix} 0.50 \\ 0.45 \\ 0.51 \end{bmatrix}, H = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } W = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.1 \end{bmatrix}$$

OR

- Q - 6 (a) Describe Least Square Estimate (LSE) and weighted LSE for estimation of x vector of n variable using another vector y of variable m where m > n. [07]
- (b) Determine the economic operating point for these three units when delivering a total of 850MW. The fuel cost functions are below; [07]

$$H_1 = 510.0 + 7.20P_1 + 0.00142P_1^2$$

$$H_2 = 310.0 + 7.85P_2 + 0.00194P_2^2$$

$$H_3 = 78.0 + 7.97P_3 + 0.00482P_3^2$$

and fuel cost functions are below Unit 1: Fuel cost=1.1Rs/MW, Unit 2: Fuel cost=1.0Rs/MW, Unit 3: Fuel cost=1.0Rs/MW. Also minimum and maximum limit of all generators are: $150 \leq P_1 \leq 600$ MW, $100 \leq P_2 \leq 400$ MW, $50 \leq P_3 \leq 200$ MW. After that use participation method to calculate the dispatch for a total load of 900MW.
